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Hierarchical temporal transformer network for tool wear state recognition

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ABSTRACT

The accurate determination of the tool-wear state helps workers maximise tool utilisation while reducing waste. It also ensures the machining quality and improves the machining efficiency. This study proposed a deep learning network based on the self-attention mechanism that enabled global modeling and long-term dependence for better tool wear state recognition by cutting signals. The Hierarchical Temporal Transformer Network (HTT-Net) was constructed by improving the Swin Transformer backbone network to enable the global modeling of the input temporal sequence signal. The token merging layer is used to build a hierarchical feature map, that enables the model to continuously increase the receptive field and improve its global modeling capability. Self-attention calculation is performed on the temporal sequence data partition window, which reduces the complexity of the model from a quadratic into a linear form of the sequence length. The shifted window enables information interaction between non-overlapping windows, which can improve the global modeling capability of the model. PHM2010 public cutting data and TC4 milling data were used for model training and test. The results showed that the tool wear state recognition capability of HTT-Net on the PHM2010 dataset outperformed the models in existing studies, with up to 16.13% improvement in recognition accuracy on the relevant dataset. The recognition accuracy on the TC4 milling dataset can reach 98.87%, which further verified the actual application capability of the model. At last, The Q-Q graph analysis verified that the model had strong robustness and stability, and the ablation experiment verified that each module of the model had positive gain on the recognition results.

1. Introduction

1.1. Background

In the actual production process, tool replacement decisions are primarily based on the operator's experience, making it troublesome to accurately determine the tool-wear state. Consequently, companies often change tools sooner than expected to avoid severe tool wear and damage to the workpiece, which can lead to significant waste. Tool-wear monitoring can reduce energy consumption and contribute to tool-health management [1,2]. Mohanraj et al. [3] suggested that the main purpose of tool-condition monitoring is the early detection of tool wear.

Based on the apparent change in slope of the tool wear curve, the tool can be categorised into three states, which are initial wear, normal wear and server wear [4]. Tool wear increases rapidly during the initial wear state, slows down during the normal wear state, and increases rapidly again during the server wear state. In the cutting process, timely replacement of tools in the server wear state can effectively ensure the

quality of machining. Therefore, it is important to analyse the signals during the machining process to determine the tool-wear state.

1.2. Related work

Manual feature extraction from machining signals by traditional machine learning algorithms for tool-condition monitoring was widely used by researchers prior to the advent of deep learning [5–7]. However, traditional machine learning algorithms require manual extraction of features, which can lead to the loss of some important information in the original signal. Moreover, traditional machine learning requires pre-processing of data through dimensionality reduction or data mapping methods. These affect the training efficiency and recognition accuracy of the model [8]. The widespread application of sensors has provided a large amount of training data for data-driven algorithms, which made deep learning widely used in industry [9]. Li et al. [10] innovatively proposed the use of large convolution kernel for aerodynamic identification to extract smoother features in the time domain and to adequately

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capture the low and medium frequency components in the frequency domain. Tang et al. [11] applied the CNN with adaptive learning rate for fault diagnosis of hydraulic piston pumps and achieved higher accuracy. Sun et al. [12] first proposed to apply transfer learning to the aerodynamic intelligence recognition task, which effectively alleviated the problem of insufficient samples in training. Deep learning algorithms can adaptively extract high-quality features from data, and have been widely used in tool-wear state recognition [13–18]. Shang et al. [13] designed DAT-Net to segment the tool wear region in the image to achieve the tool wear state recognition. Guo et al. [14] improved the interpretability of the model by multi-scale pyramid attention network. Wang et al. [16] proposed multi-label imitation learning to achieve tool wear state identification for multiple working conditions. Tool wear state recognition algorithms were continuously improved. However, the researchers did not focus on the issue of long-term dependence in cutting signals.

The signals collected during processing are generally temporal data. The long-term dependence on temporal signals has been a hot research topic in the field of deep learning. Similarly, the wear generated by a particular pass influences the subsequent tool wear. These effects are reflected in the collected cutting signals as long-term dependencies. As a result, it is indispensable to study the long-term dependence of the signal during cutting to increase the accuracy of tool-wear state recognition. Long short-term memory (LSTM) is a recurrent neural network that introduces a gate mechanism to determine the long-term dependence by controlling the circulation and loss of features. Therefore, LSTM has been well established for sequence models [19–21]. LSTM is also widely used in tool-wear monitoring [22–24]. However, LSTM cannot be parallelised during training, which has resulted in the high consumption of computational resources during the training process. In contrast, convolutional neural network (CNN) have excellent parallel computing capabilities, hence, they are more frequently used in industrial tasks [25–29]. CNN are widely used in tool-wear monitoring [30–32]. However, the increase in the CNN receptive field requires the constant increase in the depth of the CNN, which leads to degradation, gradient vanishing, and gradient explosion during training. Even if the Residual Network could solve these problems, it also increases the difficulty of network construction. Therefore, an algorithm is required that can operate in parallel while considering the long-term dependencies of cutting signals to enhance the accuracy of tool-wear recognition.

Vaswani et al. [33] proposed the transformer network, which has achieved excellent performance in machine translation. The multi-head self-attention mechanism of the transformer network can learn long-term dependencies to achieve global modelling through just one layer of the network. Transformer can also perform parallel operations, which reduces the computational cost to a certain extent. Networks based on the transformer network have demonstrated excellent performance in tasks such as natural language processing [34–36] and image classification [37–39]. The transformer network was initially applied in tool-wear prediction. Liu et al. [40] first proposed the application of the transformer in tool-wear prediction tasks, and related experiments verified its feasibility in such tasks. In order to further improve the performance of Transformer in tool-wear prediction, Liu et al. [41] further combined CNN with Transformer, and the proposed CNN-Transformer has good performance in public datasets. Agung et al. [42] added data denoising and feature selection blocks in front of the Transformer encoder to solve problems such as data redundancy, allowing the overall network to realize end-to-end tool-wear prediction. Li et al. [43] proposed the application of the informer encoder to achieve the global feature extraction of long-sequence signals in tool-wear prediction. The transformer exhibited improved tool-wear prediction results. However, it has not been applied in tool-wear state classification. Among the current limitations of the transformer is its high complexity. In comparison, the improved informer reduces the network complexity from $O(L^2)$ to $O(L\log(L))$ by sparsifying the attention matrix through

data distillation. Liu et al. [44] proposed the Swin Transformer to reduce the complexity to $O(L)$ through a window self-attention. The Swin Transformer uses a hierarchical network construction method to obtain multi-scale features. Since it has achieved excellent performance in many image classification tasks, this study converted the Swin Transformer into a 1-dimensional network and improved the related modules to enable better feature extraction for temporal data. A low-layer, low-computational cost tool wear state recognition network capable of implementing long-term dependencies and global modelling in parallel was constructed.

1.3. Contribution of this study

- (1) This study first proposed the application of a transformer-based model to the tool-wear state recognition task. Hierarchical Temporal Transformer Network (HTT-Net) was proposed for temporal data.
- (2) The Patch Merging layer and Swin Transformer block in the Swin Transformer backbone network are improved. The HTT-Net performs hierarchical network construction by token merging to continuously improve the receptive field of the model. The network complexity was reduced by performing the window self-attention operation in the non-overlapping windows of the temporal data. The shifted window allows interactions between information from non-overlapping windows, which improves the global modelling ability of the network.
- (3) The model is trained on the PHM2010 public cutting dataset and TC4 milling datasets. The results showed that the proposed model had a higher recognition accuracy compared to other tool wear state recognition algorithms, and it can also perform very well in actual applications.
- (4) The computational complexity, average inference time and training time of the HTT-Net are used to evaluate the computational cost. The computational cost is dramatically reduced compared to traditional Transformer and more advanced Transformer variants (CNN-Transformer and Informer).
- (5) The distribution of training and test datasets was analyzed by Quantile-quantile graph, and combined with the experimental results, the proposed HTT-Net model was verified to have better robustness and generalization ability as well as stability.

2. HTT-Net tool wear state recognition network framework

2.1. Main structure

In this study, we improved the Swin Transformer backbone network and designed the HTT-Net for temporal data, which was used to perform tool-wear state recognition. The main structure of the network is shown in Fig. 1. The model contains five parts: Linear Embedding, 1D Temporal Transformer Block, Token Merging, Fully connected layer, and Class. The data size of the network input was determined as the Sequence - Length $\times$ Channel. The Sequence Length dimension determines the length of the samples, while various types of signals responding to the cutting state are stacked in the Channel dimension. Linear Embedding maps the Channel dimension of the tensor to any higher dimension, allowing subsequent networks to extract additional features of different dimensions (from C0 to C1 here). Token Merging downsamples the Sequence Length dimension of the feature map by a factor of two to improve the receptive field of the network and optimise the global modelling of the model, while the Channel dimension is up-stacked by a factor of two to reduce the loss of features (Fig. 1). Fully connected layer maps the extracted features to the classification dimension Class Number (here, the number of dimensions is three) by linear mapping. The dimensions of the feature map in each layer of the network are listed in Table 1.

The 1D Temporal Transformer Block is the main feature extraction

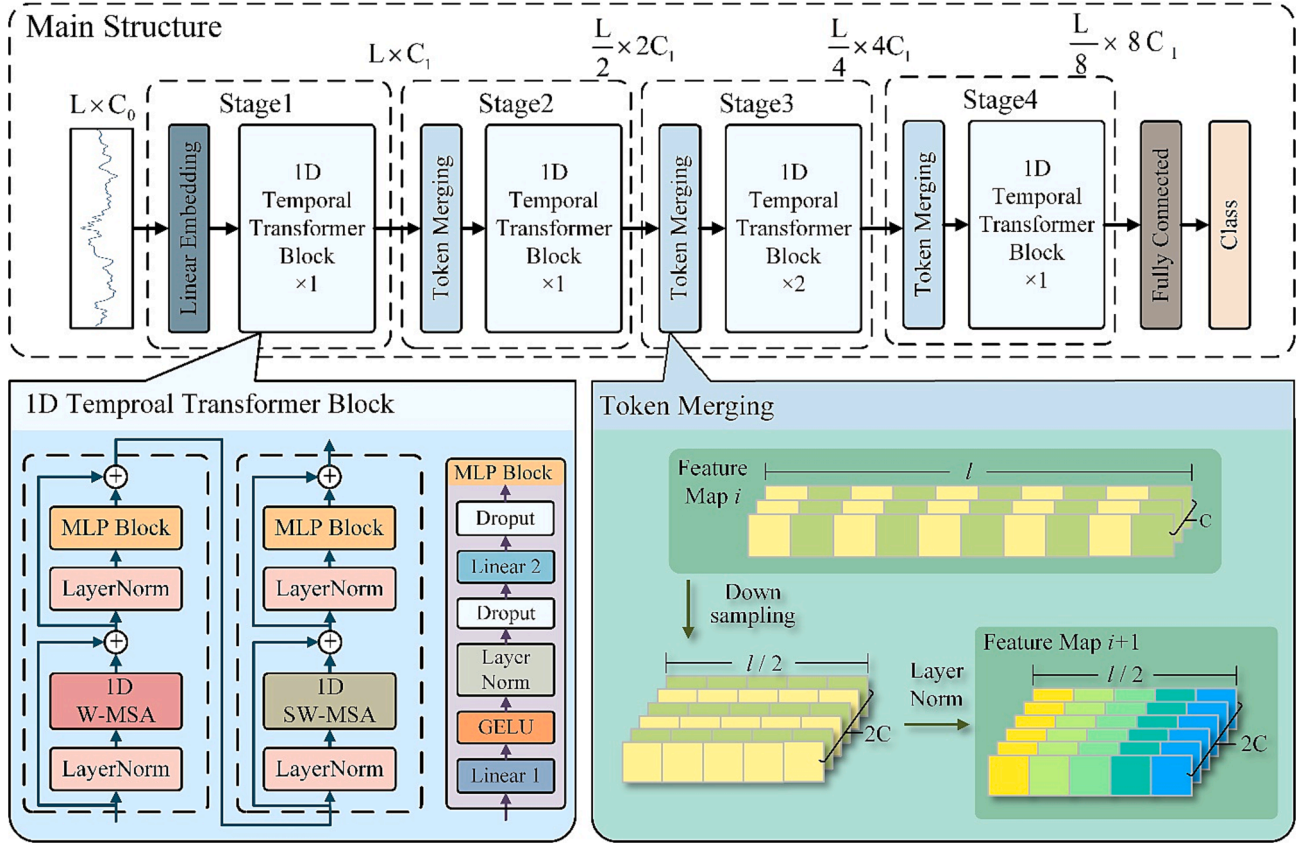


Fig. 1. The main structure of the HTT-Net and the specific structure of the 1D Temporal Transformer Block and Token Merging.

Table 1

Feature map dimensions of each layer.

Layers	Size
Input	$L \times C_0$
Stage 1	$L \times C_1$
Stage 2	$L/2 \times 2C_1$
Stage 3	$L/4 \times 4C_1$
Stage 4	$L/8 \times 8C_1$
Output	Class – Number $\times 1$

module of the entire network. It consists of the following: Layer Normalization (LayerNorm), Window Multi-Head Self-Attention (W-MSA), Shifted-Window MSA (SW-MSA), and Multi-layer Perceptron (MLP) Block (Fig. 1). To reduce the effect of the batch size on network training, Layer Normalisation was used to normalise the channel dimensions of the samples, its formula is shown in Eq. (1). The W-MSA and SW-MSA adopt the multi-head dot product self-attention mechanism in the partition windows, the formula is shown in Eq. (2). Compared with the traditional attention mechanism, the MSA mechanism allows the model to study information in independent representation subspaces, which enables it to extract richer features. The MLP Block consists of the following: Linear, GELU, Dropout, and Layer Normalisation. The Linear layer mainly implements the scaling of the data Channel layer, which enables the model to extract more dimensional features. After the data is fed into the MLP Block, Linear 1 shrinks the Channel dimension of the data by a factor of four. When the data is to be exported from the MLP Block, Linear 2 magnifies the Channel dimensions by another factor of four. Dropout randomly inactivates the neurons of the neural network to prevent overfitting during training and improve the generalisation ability of the network. GELU is a Gaussian error linear unit function that prevents gradient disappearance during the training of long-sequence samples. The complete form is shown in Eq. (3). The function in Eq.

(4) was used in the network.

$$y = \frac{x - E[x]}{\sqrt{\text{Var}[x] + \epsilon}} * \gamma + \beta \quad (1)$$

where $E[x]$ is the average value of all dimensional features from one single sample; $\text{Var}[x]$ is the variance of all dimensional features from one single sample; γ and β are the learnable parameter vectors; and ϵ is a small constant set to prevent the denominator of 0 from making the fraction meaningless, and usually set to 10^{-5} .

$$\mathbf{Q}_i = \mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}_i = \mathbf{K}\mathbf{W}_i^K, \mathbf{V}_i = \mathbf{V}\mathbf{W}_i^V,$$

$$\text{head}_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) = \text{Softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_k}}\right) \mathbf{V}_i,$$

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) \mathbf{W}^O \quad (2)$$

where h is the number of branch heads of the self-attention mechanism, $\mathbf{Q}$ is the query matrix for the whole sample, $\mathbf{K}$ is the key matrix for the whole sample, and $\mathbf{V}$ is the value matrix for the whole sample. $\mathbf{Q} \in \mathbb{R}^{L_m \times C_m}$, $\mathbf{K} \in \mathbb{R}^{L_m \times C_m}$, and $\mathbf{V} \in \mathbb{R}^{L_m \times C_m}$, where L_m is the Sequence Length dimension and C_m is the Channel dimension of the feature map. $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V, \mathbf{W}^O$ are all parameter matrices, $\mathbf{W}_i^Q \in \mathbb{R}^{C_m \times (\frac{C_m}{h})}$, $\mathbf{W}_i^K \in \mathbb{R}^{C_m \times (\frac{C_m}{h})}$, $\mathbf{W}_i^V \in \mathbb{R}^{C_m \times (\frac{C_m}{h})}$, and $\mathbf{W}^O \in \mathbb{R}^{C_m \times C_m}$. $\mathbf{Q}_i$ is the query matrix for the head $_i$, $\mathbf{K}_i$ is the key matrix for the head $_i$, and $\mathbf{V}_i$ is the value matrix for the head $_i$.

$$xP(X < x) = x \int_{-\infty}^x \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi}\sigma} dx \quad (3)$$

$$\text{GELU}(x) = 0.5x \left(1 + \tanh\left(\sqrt{2/\pi} (x + 0.044715x^3)\right) \right) \quad (4)$$

where, μ is the sample mean, σ is the sample standard deviation and $\tanh()$ is the hyperbolic tangent function.

2.2. Window self-attention and shifted window for temporal data

The complexity of the transformer is quadratic with respect to the length of the sample sequence. The temporal signal collected in industrial production is generally long, which leads to high computational costs in the application of the transformer network. Therefore, in this study, the window self-attention of the Swin Transformer was adopted for the temporal data. As shown in Fig. 2, the temporal data was divided into non-overlapping windows of length T in layer l . The operation of the self-attention is conducted in individual windows. Compared with the traditional global self-attention mechanism, the computational complexity of the overall network is reduced. The computational complexity of the traditional and window self-attention are shown in Eq. (5) and Eq. (6), respectively. The computational complexity of the traditional self-attention is quadratic with the sequence length, while that of the window self-attention mechanism is linear with the sequence length.

$$\Omega_{\text{MSA}}(L) = 4LC^2 + 2L^2C \quad (5)$$

$$\Omega_{\text{W-MSA}}(L) = 4LC^2 + 2TLC \quad (6)$$

where T is the length of the window, L is the length of the sequential sample, and C is the number of channels of the sequential sample.

However, the model cannot extract the global information of the sequence because of the non-overlapping windows, which affects its global modelling ability. Therefore, a shifted-window operation was used to realise information interaction between different windows. As shown in Fig. 2, the windows in layer $l+1$ were shifted by $\lfloor T/2 \rfloor$ time-stamps in the temporal direction to repartition the windows. Compared with the window in the l -layer, window W'_1 contains the data in windows W_1 and window W_2 of the l -layer network. The self-attention mechanism

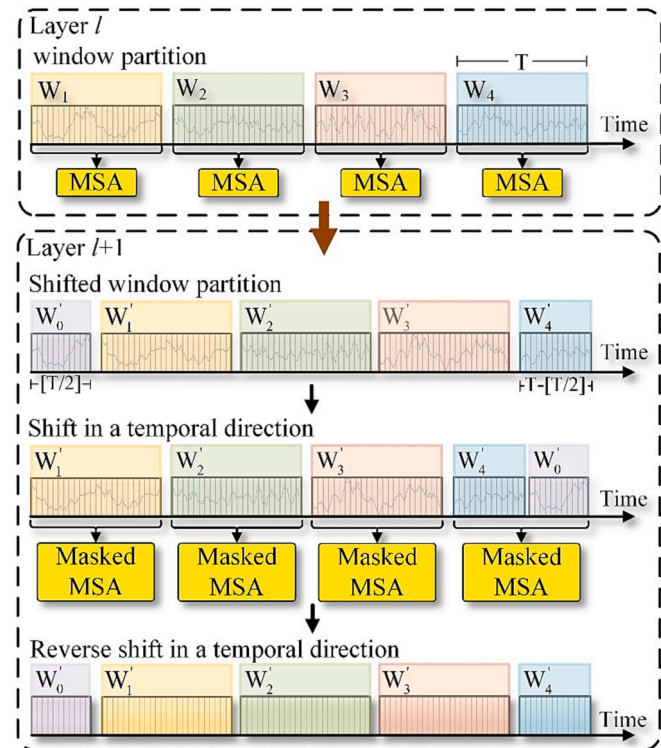


Fig. 2. Window self-attention and shifted-window self-attention.

operation on W'_1 can realise the information interaction between windows W_1 and W_2 . Therefore, the information interaction between the windows in the l -layer can be realised in the $l+1$ layer by the shifted window, which enables the model to extract the global features. However, the lengths of the repartitioned windows are inconsistent. For example, W'_0 and W'_4 (Fig. 2) are not consistent with the other window lengths, which prevents the model from running in parallel and significantly increases its complexity. Therefore, window W'_0 was shifted along the temporal direction and concatenated with window W'_4 , such that they form a new window of the same length as the other windows. Because the concatenated windows contain information from different time periods, the direct operation of the self-attention prevents the model from extracting reasonable features. Finally, window W'_0 was shifted to its original position to form the complete feature sequence.

2.3. Relative position bias

The tool-wear state during the cutting process can be defined as a function of time, which also makes the collected signal a function of time. The position of the signal has a certain temporal nature, making the extraction of position features one of the main concerns of the tool-wear recognition network. The self-attention mechanism can only determine the relationship between sample points. Therefore, additional positional encoding is required to extract the positional features by the model. This study implemented the relative position encoding of the samples by adding a bias term to the self-attention operation, as shown in Eq. (7).

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{B}\right)\mathbf{V} \quad (7)$$

where $\mathbf{Q}$ is the query matrix, $\mathbf{K}$ is the key matrix, and $\mathbf{V}$ is the value matrix. $\mathbf{B}$ is the bias matrix and the elements in $\mathbf{B}$ are the biases related to the relative positions of the samples, $\mathbf{B} \in \mathbb{R}^{L_m \times L_m}$.

For a sample t of cardinality L , there exists a relative position vector a_i between sample point t_i and other sample points.

$$t = \{t_0, t_1, t_2, \dots, t_{i-1}, t_i, t_{i+1}, \dots, t_{L-1}\} \quad (8)$$

$$a_i = (-i, -(i-1), -(i-2), \dots, -1, 0, 1, \dots, L-i-1) \quad (9)$$

The relative position vectors of all the L sample points can be arranged in rows to form a relative position matrix $\mathbf{A}$. The a_{ij} in matrix $\mathbf{A}$ represents the relative position information of sample point t_j with respect to sample point t_i . Matrix $\mathbf{A}$ has a total of $2L-1$ different element values, which represent $2L-1$ different relative position relationships. A trainable table of length $2L-1$ was set up, in which $2L-1$ relative position biases were stored for the model training. The elements in matrix $\mathbf{A}$ were used as index values to index the corresponding relative biases in the table.

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 & \dots & L-2 & L-1 \\ -1 & 0 & 1 & \dots & L-3 & L-2 \\ -2 & -1 & 0 & \dots & L & L-3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 2-L & 1-L & -L & \dots & 0 & 1 \\ 1-L & 2-L & 3-L & \dots & -1 & 0 \end{bmatrix} \quad (10)$$

where $\mathbf{A} \in \mathbb{R}^{L \times L}$ and L is the cardinality of the sample.

The presence of negative numbers in the relative position matrix $\mathbf{A}$ has made it inconvenient for computer indexing. Therefore, the relative position index matrix $\mathbf{P}$ for the whole sample was obtained by adding $L-1$ to each element in matrix $\mathbf{A}$. According to each element in the relative position index matrix $\mathbf{P}$, the corresponding biases were selected in the table for replacement to form the bias matrix $\mathbf{B}$.

$$\mathbf{P} = \begin{bmatrix} L-1 & L & L+1 & \dots & 2L-3 & 2L-2 \\ L-2 & L-1 & L & \dots & 2L-4 & 2L-3 \\ L-3 & L-2 & L-1 & \dots & 2L-5 & 2L-4 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 2 & 3 & \dots & L-1 & L \\ 0 & 1 & 2 & \dots & L-2 & L-1 \end{bmatrix} \quad (11)$$

where $\mathbf{P} \in \mathbb{R}^{L \times L}$ and L is the cardinality of the sample.

3. Experiment study

3.1. Data description

3.1.1. Benchmark dataset

In order to compare the performance of the proposed HTT-Net model under the same conditions, The PHM2010 tool-wear public dataset was used in this experiment [45]. The dataset contains six tool lifecycle experiments (C1 ~ C6). Among these, three cutting experiments (C1, C4, and C6) recorded the tool wear values for each pass. In this experiment, the Inconel 718 alloy was dry-milled with a three-tooth carbide ball-end milling cutter (Fig. 3). The experimental parameters are listed in Table 2.

The milling experiment is side milling. The length of each pass was 108 mm. During cutting, three types of sensors were used to collect the three-axis force, three-axis vibration, and acoustic emission signals (i.e. a total of seven signals) online. An NI DAQ PCI 1200 signal acquisition card was used for the seven channel signals acquisition with a sampling frequency of 50 kHz. The width of the flank wear land VB on the flank face was used as the wear value of the tool. At the end of each pass, a LEICA MZ12 microscope was used to measure the flank wear of the three cutter teeth offline. In order to intuitively illustrate the areas of tool wear measured in the dataset and the different stages of tool wear, several pictures were taken in one of the factories, as shown in the Off-line measurement section of Fig. 3. The mean of the three flank-wear values was selected as the tool-wear value. The wear curves of the C1, C4, and C6 tool lifecycles are shown in Fig. 4.

3.1.2. TC4 milling experimental data

In order to subsequently better validate the ability of the proposed model to be applied in actual tests, a full life cycle experiment of a tool milling TC4 titanium alloy was conducted (Fig. 5a). The experimental machine tool was a Ningqing VC1060 3-axis milling machine. The dimensions of the workpiece were $100 \times 80 \times 60$ mm. Experiments were

Table 2

PHM2010 cutting experiment parameters.

Experiment parameters	Values
Spindle speed (n)	10400 rpm
Feed per turn (f)	0.15 mm/r
Cutting depths (a_p)	0.2 mm
Cutting width (a_e)	0.125 mm
Sampling frequency	50 kHz

performed with dry milling and side milling. The milling tool head was F4042.W32.032.Z03.11 high-performance 3-tooth milling cutter from Walter, and the inserts were ADHT 1204PER-G88 WK10 indexable inserts from Walter. The Kistler 9265B dynamometer was used for force signal monitoring during milling. The signal acquisition card was Measurement Computing's pcm-das16jr-16. The collected force signal is shown in Fig. 6 after compensation for zero-point drift. The experimental parameters are shown in Table 3. An optical microscope was used to measure width of flank wear land VB after each pass (Fig. 5b), for a total of 65 passes. Each pass length was 100 mm. The mean of the three flank-wear values was selected as the tool-wear value. The wear curve for the full lifecycle of the tool and different stages of the flank wear are shown in Fig. 7.

3.2. Datasets preparation

In PHM2010 datasets, the tools in all three experiments almost entered the normal wear at the 50th pass. However, the length of time to enter the severe wear stage was inconsistent. Considering that the tool wear in experiment C6 was consistently higher than that in the other two experiments, the time when the tool entered the severe wear in experiment C6 was used as a reference to classify the normal and severe wear. Therefore, the tool wear from the 1st to 50th pass was set as the initial wear, 51st to 175th pass as normal wear, and 176th to 315th pass as severe wear (Fig. 4). The force signals from experiments C1, C4, and C6 on the PHM2010 dataset were selected as the input data of the model.

A huge amount of data was generated by each pass, which means that the direct use of data from a single pass as a sample would lead to highly complex calculations and overfitting. To solve this problem and increase the size of the training set, the data from each pass was divided into several small samples of size 2000×3 using a non-overlapping window of length 2000. Because the data size in the three stages was unbalanced, especially when the size of the initial wear data was much smaller than that of the normal and severe wear, the force signals of the

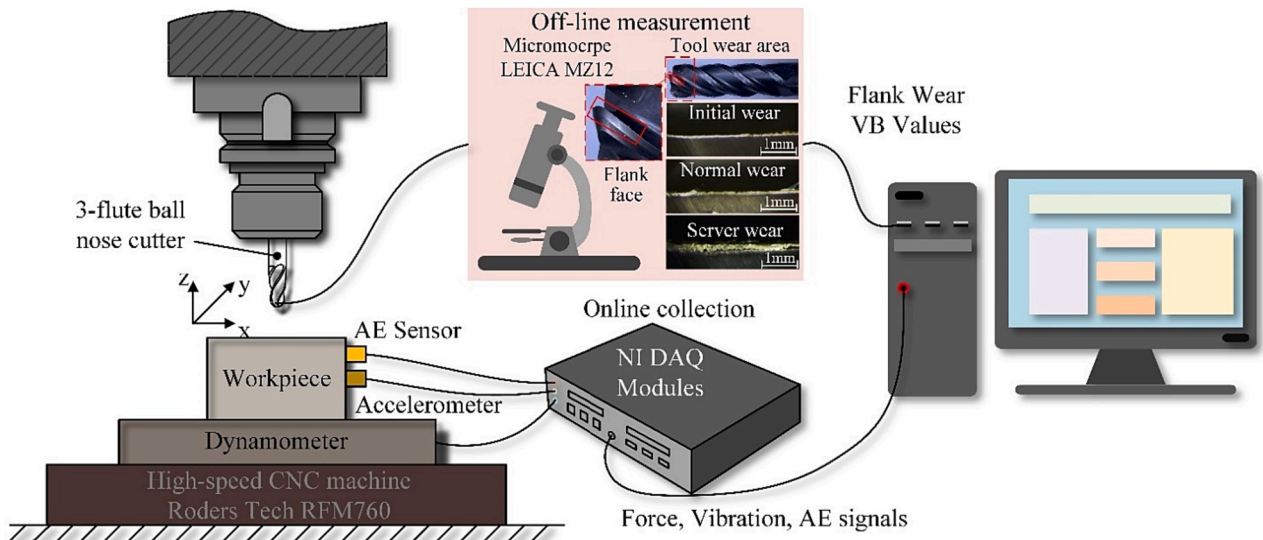


Fig. 3. Schematic diagram of the PHM2010 experiment.

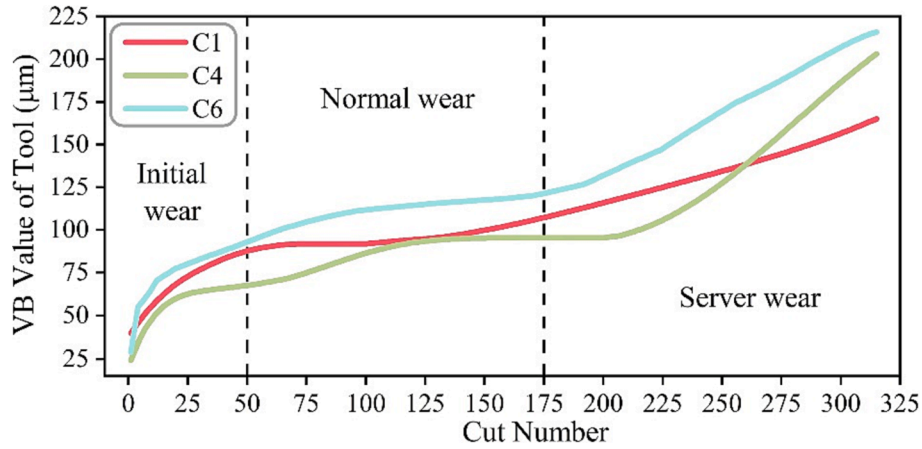


Fig. 4. Tool lifecycle wear curves of PHM2010 datasets.

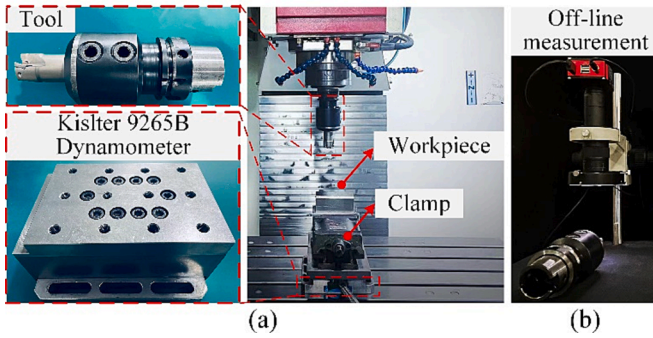


Fig. 5. TC4 milling experiment (a) Milling and signal acquisition devices (b) Tool wear measurement device.

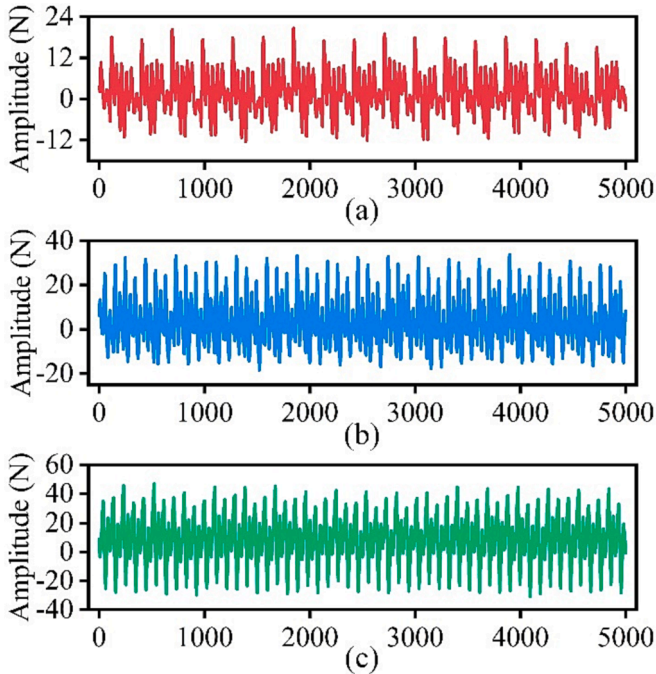


Fig. 6. Amplitude of force signals (y direction) for three wear states in TC4 milling experiment (a) Initial wear (b) Normal wear (c) Severe wear.

Table 3

TC4 milling experiment parameters.

Experiment parameters	Values
Spindle speed (n)	1500 rpm
Feed per turn (f)	0.3 mm/r
Cutting depths (a_p)	4 mm
Cutting width (a_e)	1 mm
Sampling frequency	1.5 kHz

initial wear were re-sampled by the partition window from a new starting point to upsample the data. Meanwhile, the normal and severe wear data were randomly downsampled. Finally, the experimental data from C1, C4, and C6 were converted into a three-class dataset containing 10,000 samples for each class. To improve the evaluation of the model performance, two types of crossover datasets were set up (M and T datasets). The detailed settings of the datasets are listed in Table 4 and Table 5.

The TC4 milling experiment data was also divided into initial wear, normal wear and severe wear according to the tool wear curve. The width of flank wear land VB increased rapidly from the 1st to the 15th passes, and this state was defined as initial wear. The width of flank wear land VB increased slowly from the 16th to the 43rd passes, and this state was defined as normal wear. After the 44th pass the tool wear increased rapidly again and this state was defined as severe wear. The quantity of each class was balanced by upsampling. There were 2000 samples in each class, and 6000 samples in total. 80 % of the samples were randomly selected as the training dataset and the remaining 20 % as the test dataset.

3.3. Model training and test

3.3.1. Training configuration

The training host was equipped with the Win10 system. The training environment was configured with PyTorch 1.10. The model was trained in an NVIDIA RTX3060 GPU with graphics driver version 497.29 and CUDA version 11.5. Owing to the large amount of data, a mini-batch training strategy was used. The AdamW optimiser was used for iterative optimisation of model parameters. The weight decay parameter in the optimiser was set to reduce the overfitting of the network. The hyperparameters of the network settings are listed in Table 6.

3.3.2. Model training and test intermediate results

In order to better describe the training process of the proposed model, the loss curves of the training dataset and the accuracy curves of the test dataset during model training are plotted (Fig. 8). It can be seen that the model converged before Epoch 5 during training, which was a

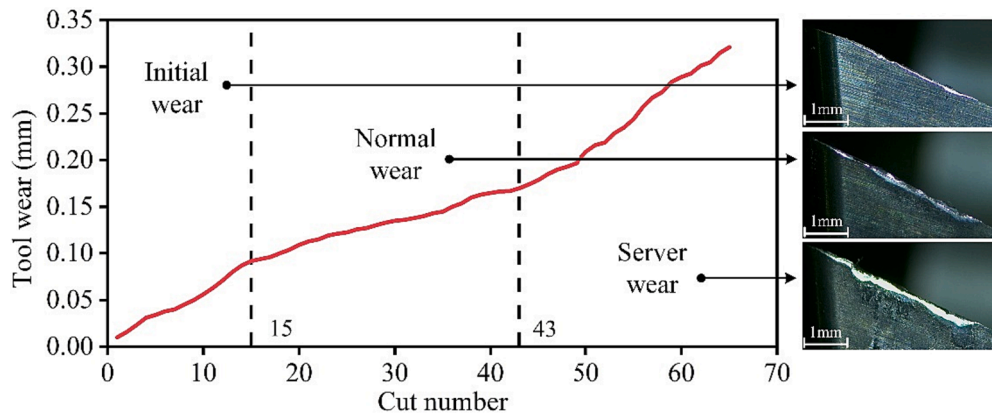


Fig. 7. Tool lifecycle wear curves of TC4 milling experiment.

Table 4

The specific settings of the M dataset.

Datasets	Training datasets	Test datasets 1	Test datasets 2
M1	C1	C4 (Recorded as M11)	C6 (Recorded as M12)
M2	C4	C1 (Recorded as M21)	C6 (Recorded as M22)
M3	C6	C1 (Recorded as M31)	C4 (Recorded as M32)

Table 5

The specific settings of the T dataset.

Datasets	Training datasets	Test datasets
T1	C1-C4	C6
T2	C4-C6	C1
T3	C6-C1	C4

Table 6

Setting of hyperparameters.

Parameters	Values
Learning rates	0.0001
Batch size	16
Weight decay	0.05

very fast training process. The test dataset also achieved high accuracy very quickly, which showed that the model can learn the common features under the training and test datasets very well.

3.3.3. Model performance evaluation

Benchmark dataset evaluation

A confusion matrix was used to present the accuracy of the model in different categories of the M and T dataset (Fig. 9 and Fig. 10). To further explore the model performance, four secondary indicators of the confusion matrix were used, respectively, accuracy, precision, recall, and specificity. Accuracy measures the overall classification ability of the model. Precision measures the accuracy of the model in making recognitions for each class. Recall can measure the completeness of the model for each type of tool wear state. Specificity measures the model's misclassification rate for each category. In the tool wear state recognition task, the initial and normal wear stage needs to reduce false alarms to avoid premature tool change leading to waste. While the severe wear stage should focus on checking all the samples to avoid too late tool change resulting in lower machining quality. Therefore, more attention should be paid to precision in the initial and normal wear stages, while recall and specificity should be given more attention in the severe wear stages. The values of the indicators for the M and T datasets are shown in Table 7 and Table 8. It can be seen that the average precision in both

datasets in the initial and normal wear stages reached 93.98 % and 90.97 %, and the average recall and specificity in the severe wear stage reached 95.50 % and 97.80 %.

ROC curve provides a comprehensive evaluation of the model. The closer the curve is to the top-left corner, the better the performance of the classifier. The area under the ROC curve (AUC) is another evaluation metric that can assess the overall performance of the model. The ROC curves for the M and T datasets are depicted in Fig. 11, and the corresponding AUC values are shown in Table 7 and Table 8. It can be seen that the AUC of the proposed model exceeded 95 % on both M and T datasets.

3.3.4. Actual application evaluation

In order to investigate the capability of the proposed model in actual applications, the HTT-Net model was evaluated for tool wear recognition results on the TC4 milling dataset by means of confusion matrices (Fig. 12a). The four secondary indicators of the confusion matrix were also used to further evaluate the actual application capability of the proposed model. The secondary indicators are shown in Table 9. The high precision of the proposed model in the initial wear and normal wear stages indicated that the proposed model had a low false alarm rate for the tool in these two wear states in actual application, which could avoid misleading the operator to change the tool prematurely. The proposed model also had high recall and specificity in the severe wear stage, which indicated that the HTT-Net model can be well recognized for severe wear tools in actual applications, which can avoid too late tool change that could lead to degradation of the machining quality of the workpieces. To further evaluate the performance of the proposed model for actual applications, the ROC curves of the model on the TC4 milling dataset (Fig. 12b) were plotted and the AUC was calculated. The values of AUC are shown in Table 9. It can be seen that the ROC curve tended to the upper left corner and the AUC was 0.9992, which is very close to one. This indicated that the model performed well on the actual dataset. These positive results demonstrated the superior performance of the proposed HTT-Net model when the tool wear state recognition task is actually performed.

3.4. Comparison of experimental results

3.4.1. Accuracy comparison.

Accuracy was used by most researchers to characterize the performance of the model. Therefore, accuracy was also used as a criterion for comparison with other studies. Only very few researchers used the M dataset for the validation of tool wear state recognition algorithms. It was known from the study in Section 3.5 that the difference between the training and test datasets distributions for the M dataset was larger than that for the T dataset. Therefore, the M dataset required higher generalization ability and robustness of the model. In order to validate the

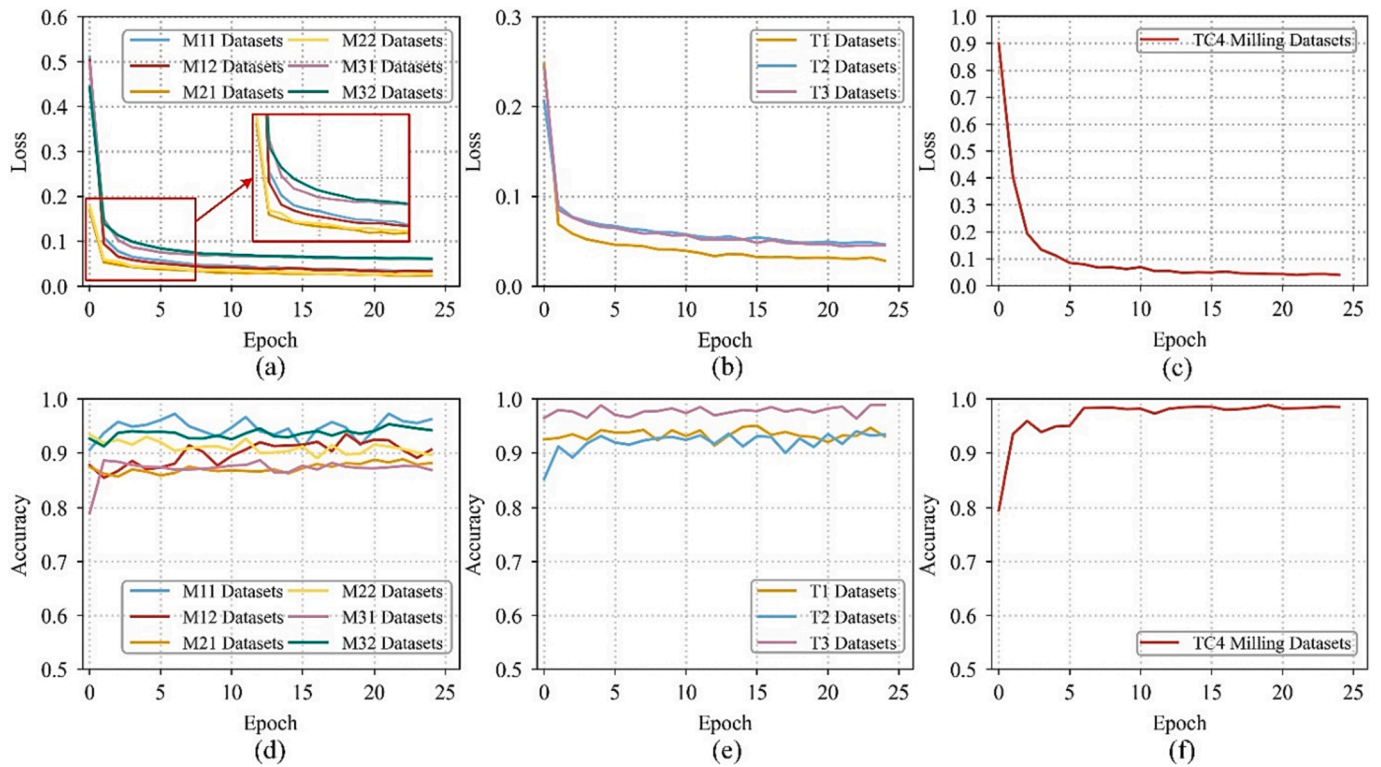


Fig. 8. Loss curves on the training dataset and Accuracy curves on the test dataset of the proposed model (a) Loss curves for the training datasets of the proposed model on the M datasets, (b) T datasets, and (c) TC4 milling dataset; (d) Accuracy curves for the test datasets of the proposed model on the M datasets, (e) T datasets, and (f) TC4 milling dataset.

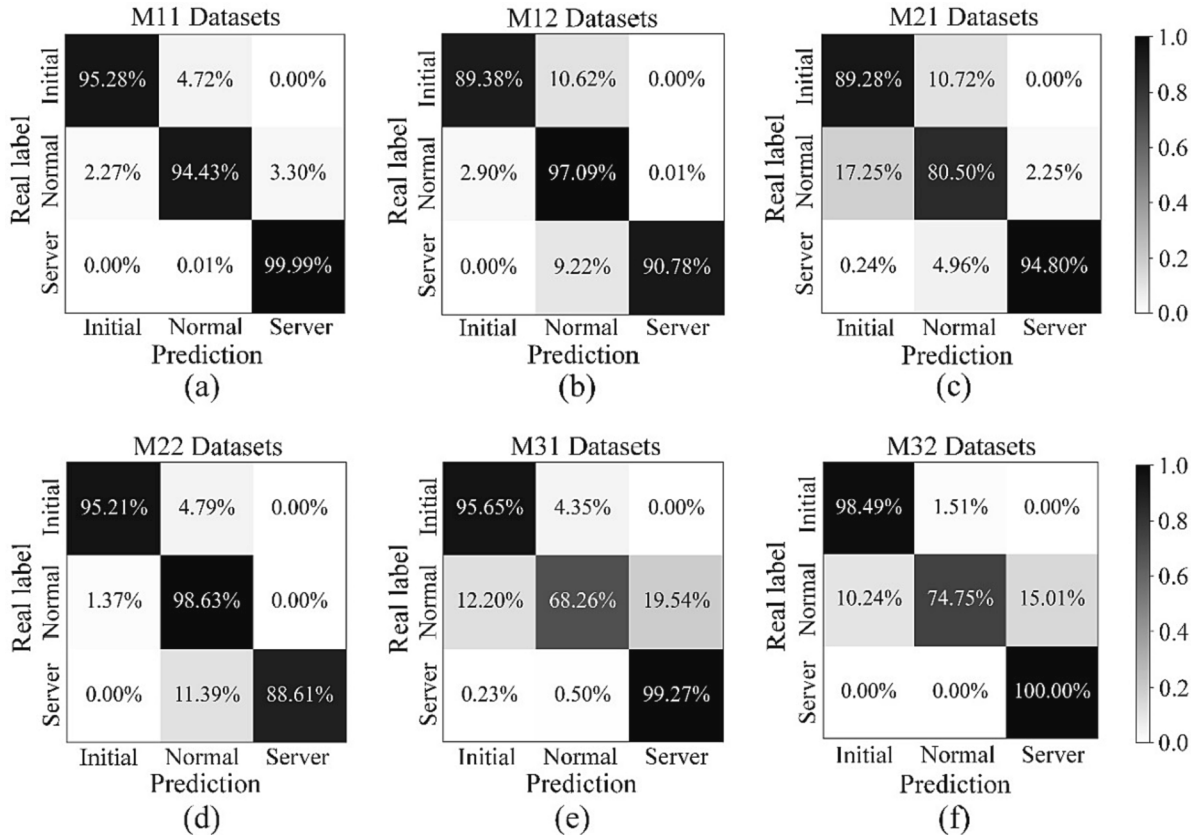


Fig. 9. Confusion matrix of experimental results on the M dataset (a) C1 test C4, (b) C1 test C6, (c) C4 test C1, (d) C4 test C6, (e) C6 test C1, and (e) C6 test C4.

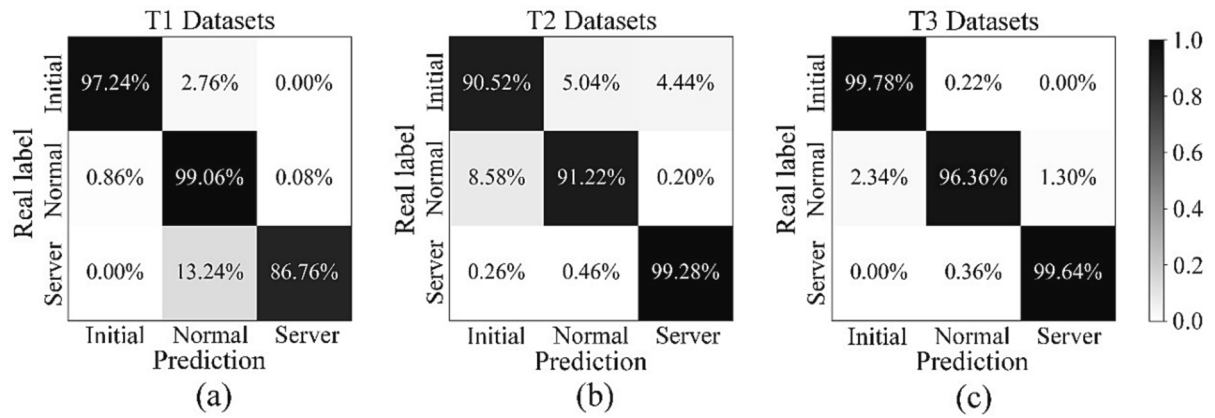


Fig. 10. Confusion matrix of experimental results on T dataset (a) C1 and C4 test C6, (b) C4 and C6 test C1, (c) C6 and C1 test C4.

Table 7

Evaluation indicators on the M dataset.

Evaluation indicators		M11	M12	M21	M22	M31	M32
Accuracy (%)		96.57	92.42	88.19	94.14	87.73	91.08
	P _{initial}	97.67	96.86	83.62	98.58	91.26	90.58
Precision (%)	P _{normal}	95.23	83.03	83.70	85.91	93.37	98.02
	R _{server}	99.99	90.78	94.90	88.61	99.27	100.00
Recall (%)	S _{server}	99.99	95.59	97.44	94.41	99.60	100.00
Specificity (%)		0.9978	0.9888	0.9616	0.9874	0.9520	0.9899
AUC							

Table 8

Evaluation indicators on the T dataset.

Indicators		T1	T2	T3
Accuracy (%)		94.35	93.67	98.59
	P _{initial}	99.12	91.10	97.08
Precision (%)	P _{normal}	86.09	94.31	99.40
	R _{server}	86.76	99.28	99.64
Recall (%)	S _{server}	93.79	99.53	99.82
Specificity (%)		0.9790	0.9770	0.9996
AUC				

performance of the proposed model on the M dataset, the only model in the existing paper that used the M dataset was used for comparison. The comparison model adopted the signal imaging method to convert the

temporal sequence data into two-dimensional images, and realised the classification of tool-wear states through two-dimensional convolution feature extraction [46]. In comparison, the HTT-Net improved the classification accuracy of each category by more than 6 %. In particular, the classification accuracy of the HTT-Net on the M31 dataset increased by 16.13 %. The comparison of the classification accuracies of the two models is presented in Fig. 13.

More researchers selected the T dataset to validate the model. To validate the performance of the models on the T dataset, models from existing papers, which were validated using the T model, were used for comparison. The mixed model based on a 1D CNN and deep generalised canonical correlation analysis [30] exhibited the highest accuracy of 89.5 % on the T1 dataset. The mixed model based on a gradient-enhanced decision tree and mixed-classification restricted Boltzmann

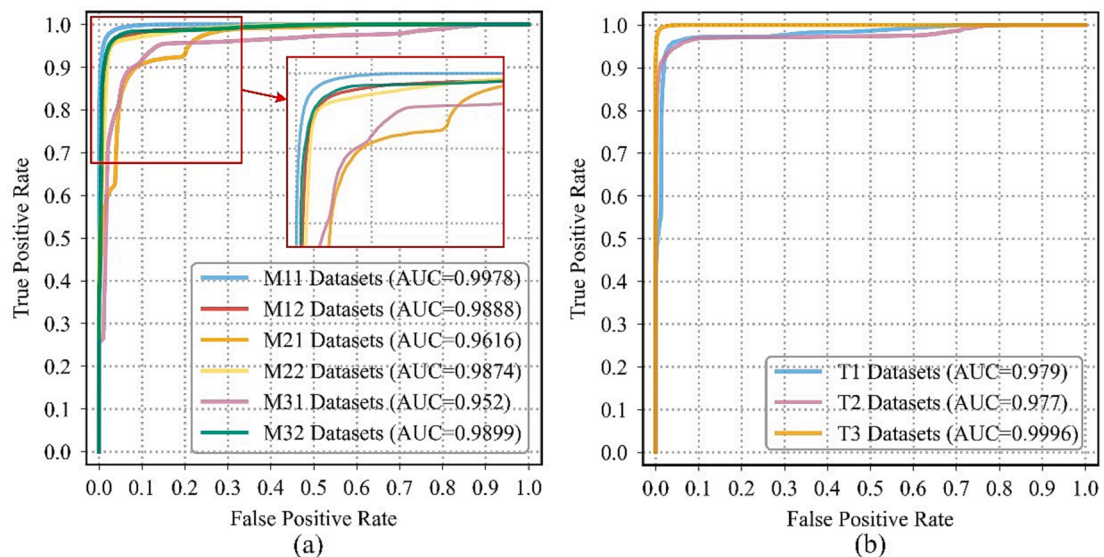


Fig. 11. ROC curves for the M and T datasets (a) ROC curve for the M dataset, (b) ROC curve for the T dataset.

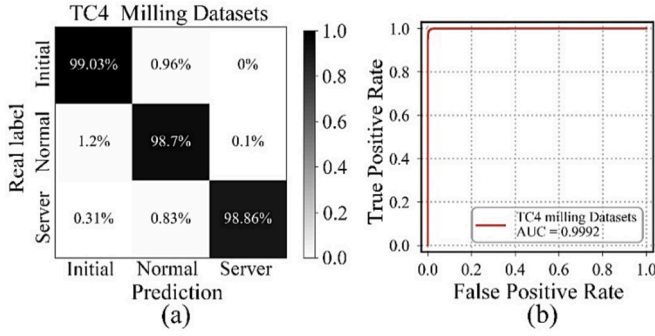


Fig. 12. TC4 milling dataset confusion matrix and ROC curve (a) Confusion matrix (b) ROC curve.

Table 9

Evaluation indicators on the TC4 Milling dataset.

Indicators		TC4 milling dataset
Accuracy (%)		98.87
Precision (%)	$P_{initial}$	98.50
	P_{normal}	99.90
Recall (%)	R_{server}	98.86
Specificity (%)	S_{server}	99.42
AUC		0.9992

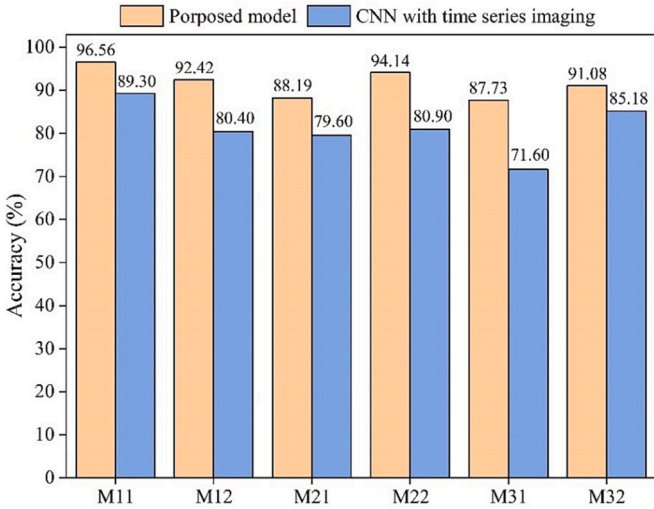


Fig. 13. Comparison of the accuracy of the proposed model with the only existing model on the M datasets.

machine [47] achieved the highest accuracy of 90.8 % on the T2 dataset. The mixed model based on the CaAt1 channel attention mechanism and ResNet-1d [48] achieved the highest accuracy of 89.06 % on the T3 dataset. To be able to compare with more deep learning models, three tool wear state recognition models were built based on GRU [49], LSTM [50], Transformer [33], CNN-Transformer [31] and Informer [43] model principles. The hidden layer dimension of GRU and LSTM was set to 32 and the number of hidden layers was set to two (Fig. 14a). The Transformer model stacked the Encoder layer of the traditional Transformer 10 times (Fig. 14b), where the number of heads of multi-head attention was set to three. The feed forward layer had the same structure and parameter settings as the proposed HTT-Net's MLP Block. The CNN-Transformer model consisted of a parallel CNN part and a Transformer part (Fig. 14c). The Transformer part was consistent with the Transformer model built above. The CNN part was designed with a total of three layers, each with a convolutional kernel size of three, padding

size of one, and max pooling kernel size of two. The first layer composed of Conv1d, Batch Normalization and Max pooling, here the output dimension of Conv1d was set to eight. The second layer had the same structure as the first, with the output dimension of Conv1d was set to 16. The third layer consisted of three Conv1d with output sizes of 32, 32, and 64. The features extracted from the two parts were stitched into new features after Flatten layer and finally mapped to the output layer through Fully connected layer. The Informer model was stacked eight times with Informer's Encoder layer (Fig. 14d), where the number of heads for the Multi-Head ProbSparse Self-Attention was set to three. The structure of Feed Forward was consistent with the MLP block in HTT-Net, but the two Linear layers were replaced with two Conv1d. The first Conv1d had the kernel size set to one, stride set to one, padding set to one, and the output dimension set to four times the input dimension. The second Conv1d has the kernel size set to one, stride set to one, padding set to one, and the output dimension set to one quarter of the input dimension. The kernel size of Conv1d in the Distilling layer was set to three, stride was set to one, padding was set to one, and the output dimension was three; the kernel size of max pooling was set to three, stride was set to two, and padding was set to one. A comparison of the classification accuracies of T dataset is presented in Table 10.

The classification accuracy of the HTT-Net network proposed in this study reached 94.35 %, 93.67 %, and 98.59 % on the T1, T2, and T3 datasets, which is 4.85 %, 2.87 %, 6.29 % higher than that of the best existing models, respectively. GHCRBM requires optimal feature set selection by manual selection, while HTT-Net can continuously iterate higher quality features by back-propagation when the amount of data is more sufficient. GRU and LSTM alleviate the long-term dependency problem by serial hidden layers. However, the 2000 input sequence length in this study was too long for GRU and LSTM, which led to poor performance. 1D-CNN-DGCCA and CaAt-ResNet-1d were essentially 1D-CNN applied in this task. Facing large-scale T datasets with 20,000 samples per training dataset, the transformer-based HTT-Net can extract better quality features compared to 1D-CNN. However, the conventional Transformer was also less effective than HTT-Net; a reason is only three channel layers in the input of this study, which was difficult to partition with the multi-head self-attention (MSA) of conventional Transformer. In contrast, Token Merging of HTT-Net constantly downsampled and stacked Channels, which allowed MSA to focus on a wider variety of features. As a composite model, here CNN-Transformer only fused the classification results of CNN and Transformer by weighting them with learnable weights. The final classification result was better than Transformer but still lower than the proposed HTT-Net. Moreover, the computational cost of CNN-Transformer was higher than Transformer, which affected the practical application of the model. The probspare self-attention adopted by Informer (only the encoder was used here) can also reduce the computation to a certain extent by data distillation. Compared with Transformer and CNN-Transformer, the recognition accuracy was improved and a superior recognition accuracy was achieved in the T3 dataset. However, there was still a certain gap with the proposed HTT-Net.

3.4.2. Computational cost comparison

To verify that the computational cost of the proposed model was reduced compared to the conventional Transformer and state-of-the-art variant Transformer networks (CNN-Transformer and Informer), the computational complexity (FLOPs) (Fig. 15a), average inference time (Fig. 15b), and training time (Fig. 15c) were calculated. Compared with three kinds of model (Transformer, CNN-Transformer and Informer), the FLOPs of HTT-Net were reduced by 63.10 %, 67.52 % and 29.63 % respectively. This indicated that the proposed HTT-Net had lower GPU requirements during training. The average inference time of the model refers to the time required for each iteration on the GPU. The training time of the model refers to the time required for each batch during training, which includes the time for the CPU to organize the batch input to the network. Compared with three kinds of models, the average

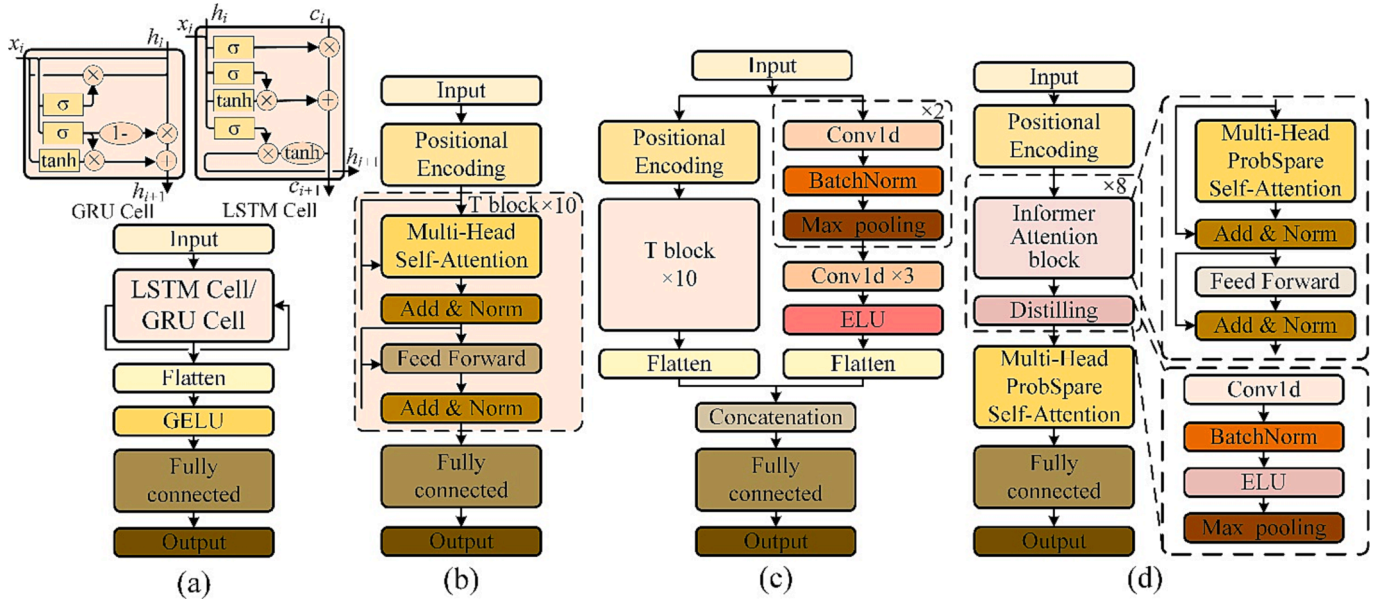


Fig. 14. Structure of different comparison models (a) LSTM and GRU (b) Transformer (c) CNN-Transformer (d) Informer.

Table 10

Comparison of experimental results of T datasets.

Models	Dataset T1 (%)	Dataset T2 (%)	Dataset T3 (%)
GHCRBM [47]	89.20	90.80	58.40
1D-CNN with DGCCA [30]	89.50	90.00	89.20
CaAt-ResNet-1d [48]	85.25	87.98	89.27
GRU [49]	82.36	84.44	82.38
LSTM [50]	82.15	81.24	78.81
Transformer [33]	89.21	84.23	88.08
CNN-Transformer [41]	88.54	85.44	90.30
Informer [43]	90.04	90.23	92.30
Proposed model	94.35 (+4.85)	93.67 (+2.87)	98.59 (+6.29)

inference time were reduced by 44.84 %, 47.62 % and 25.45 % respectively, and the training time were reduced by 24.81 %, 30.52 % and 14.70 % respectively.

3.5. Robustness and stability analysis

Quantile-quantile graphs (Q-Q graphs) (Fig. 16) were drawn to compare the distribution differences between the test and training datasets, this is significant for studying the robustness and stability of the model. The y-axis force signals were used to analyse the distribution differences between datasets. In the M dataset, the distribution differences between C4 and C6 were similar (Fig. 16b and 16c), while those between C1 and the other two datasets were different (Fig. 16a-c). Therefore, the stability of the model can be evaluated by observing the classification results when C4 and C6 were alternately training and test datasets. The robustness of the model can be evaluated through the classification results when C1 and the other two datasets were

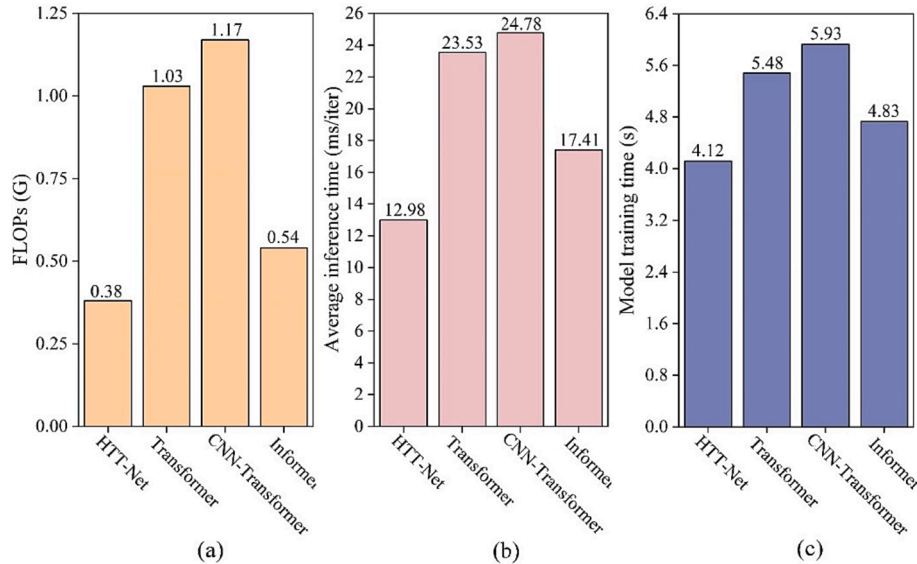


Fig. 15. Comparison of HTTP-Net and Transformer-based networks computational costs (a) Computational complexity (b) Model average inference time (c) Model training time.

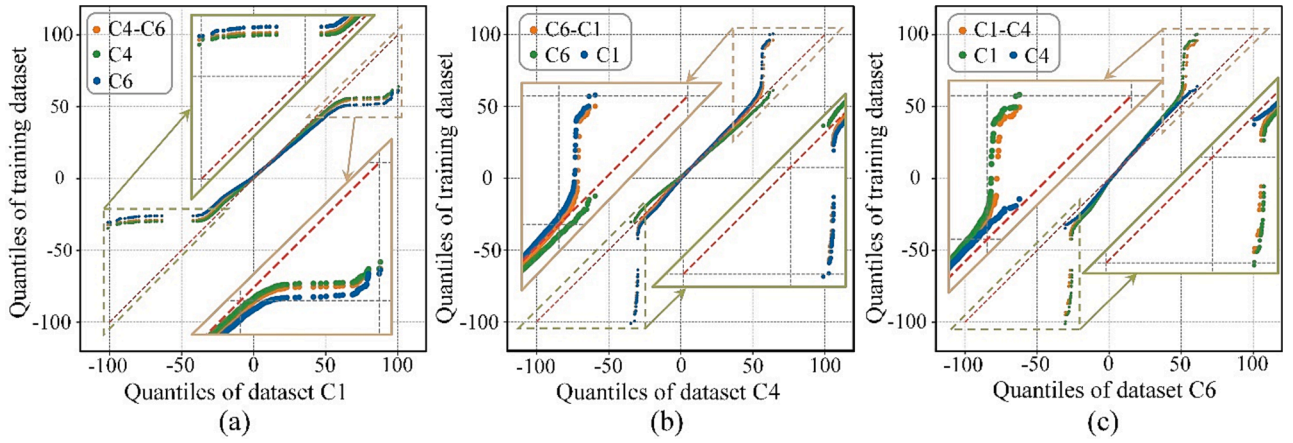


Fig. 16. Quantile-quantile graph of training and test datasets (a) C1 is the testing set, (b) C4 is testing set, and (c) C6 is the testing set.

alternately the training and test datasets. The data distribution was also reflected in the tool-wear curve. The trends in the tool-wear curves of the C4 and C6 experiments were similar, while that of C1 differed significantly from the other two experiments (Fig. 4). This also indicated a mapping relationship between the force signal during cutting and the tool-wear state. The distribution gap between the training and test datasets in the T dataset was smaller than that between C1 and the other two datasets in the M-dataset, but larger than that between the C4 and C6 datasets. Therefore, the T dataset had lower performance requirements for the classification model than the M dataset. The T dataset was selected because it can be compared with the tool-wear classification models in more studies under the same dataset.

The average classification accuracy of the mutual training and test datasets of C4 and C6 was 94.12 % (C4 test C6) and 91.08 % (C6 test C4), respectively. It can be proved that HTT-Net had good stability combined with the analysis of Q-Q graphs. The average classification accuracies of the alternate training and test datasets of C1 and the other two datasets were 96.57 % (C1 test C4), 92.42 % (C1 test C6), 88.19 % (C4 test C1), and 87.73 % (C6 test C1). It can be proved that HTT-Net had excellent robustness combined with the analysis of Q-Q graphs.

3.6. Ablation experiments

To verify the contribution of each module to the model performance, five models (Model 1, Model 2, Model 3, Model 4, and Model 5) were designed for ablation experiments. The shifted-window module was eliminated in Model 1. The relative position bias module was eliminated in Model 2. In Model 3, learnable absolute position encoding replaced the relative position bias of the HTT-Net. In Model 4, relative position encoding in the HTT-Net was replaced by relative position encoding in the XLNET style. Model 5 was the baseline network, which was a 1-dimensional network directly transformed from Swin Transformer. The T-dataset was used to evaluate the model performance experimentally. The experimental results are summarised in Table 11.

In the ablation experiment, the classification accuracy of all models decreased after the relevant modules were eliminated, which indicates the positive correlation between the designed modules and HTT-Net

performance. The elimination of the shifted window in Model 1 caused the model to perform only the self-attention operation in its own non-overlapping windows. This means that no interactions occurred between the information in each window, which affected the global modelling ability of the model. In contrast to the complete HTT-Net model, the average classification accuracy of Model 1 on the T-dataset decreased by 1.11 %. The elimination of the relative position bias in Model 2 showed that the self-attention mechanism could only obtain the weight relationship between different points in the sequence, and could not extract the temporal features of the sequence, which reduced the classification accuracy. In particular, the classification accuracy on the T2 dataset was only 85 %, which was 7.94 % lower than that of the complete HTT-Net model. In Model 3, learnable absolute position encoding was added before the data was input into the network. This enabled the model to autonomously study the absolute position feature of the sequence during training. In Model 4, XLNet-style relative position encoding was referenced. Through the trigonometric functional encoding of the relative position information of the Query matrix in self-attention, the model could obtain the corresponding temporal features during training. Compared with Model 2 without any position encoding, Models 3 and 4 exhibited improved accuracy on the T2 and T3 datasets. This indicated that position coding has a certain significance for studies on tool-wear state classification. In contrast to the proposed HTT-Net model, the average classification accuracies of Models 3 and 4 decreased by 1.44 % and 1.85 %, respectively. Since the Swin Transformer was directly transformed into a 1-dimensional network, Patch Merging and Relative position bias of model 5 still targeted the extraction of 2-dimensional features without considering the specificity of 1-dimensional temporal features. Compared with the proposed HTT-Net, the average classification accuracy of the T dataset was reduced by 1.72 %.

4. Conclusion and future work

This study proposed the first application of a transformer-based in tool-wear recognition. The HTT-Net model was developed to classify the temporal data, which was a temporal improvement of the Swin Transformer backbone network. The details of this study are as follows:

- (1) Through the token merging of feature maps, hierarchical features were obtained, while the receptive field of the network was continuously improved. The MSA was calculated in each non-overlapping window partition. The W-MSA reduced the complexity of the model from quadratic to linear with the sequence length. The shifted window was used to realise the information interaction between non-overlapping windows to improve the global modelling ability of the network. A relative

Table 11
Ablation experiment results.

Models	Dataset T1 (%)	Dataset T2 (%)	Dataset T3 (%)
Model 1 (no shifted)	93.44	92.69	97.16
Model 2 (no Relative bias)	93.83	85.73	96.71
Model 3 (Absolute position)	92.81	90.55	97.93
Model 4 (Relative position)	91.25	92.97	96.84
Model 5 (baseline network)	92.45	90.89	98.08
Proposed model	94.35	93.67	98.59

position bias was added to the self-attention to extract the relative position features of the force signal, which allowed the model to extract temporal features.

- (2) The computational cost of the proposed HTT-Net was significantly reduced compared to Transformer and the other two state-of-the-art variant network (CNN-Transformer and Informer). Compared to these three models, the overall computational complexity of the proposed HTT-Net was reduced by 63.10 %, 67.52 %, 29.63 %, respectively, which proved that the proposed model requires lower GPU performance. The reasoning speed was reduced by 44.84 %, 47.62 %, and 25.45 % compared to the other three models, and the training time of the proposed model was also reduced by 24.81 %, 30.52 %, and 14.70 %, respectively, which proved that the model had lower GPU usage time during training.
- (3) The ablation experiment was performed. The results showed that SW-MSA reduced the model complexity and improved the classification accuracy. It was also proved that the learnable relative position bias in the HTT-Net model was better than traditional relative position coding and absolute position coding for the temporal data in the tool-wear state recognition task. The comparison results with the baseline network demonstrated that the modification of the HTT-Net model with Token merging and relative position bias for 1-dimensional temporal features was useful to improve the model recognition accuracy.
- (4) Q-Q diagram was used to analyse the distribution difference between training and test datasets. HTT-Net had high recognition accuracy in training sets with close distribution differences, which verified the stability of the model. HTT-Net also had high recognition accuracy in data sets with large distribution differences, which verified the robustness of the model.
- (5) The tool-wear classification accuracy of the HTT-Net network on the PHM2010 public cutting data was higher than that of the models proposed in previous studies. On the T dataset, the classification accuracy of the T3 dataset upgraded by 6.29 % contrasted with the top-performing existing model. On the M dataset, the average classification accuracy of the three cross-datasets reached 91.69 %. The classification accuracy of the M3 dataset was 16.13 % higher than that of the top-performing existing model. The proposed HTT-Net model achieved 98.87 % recognition accuracy on the real TC4 milling tool wear dataset, which confirmed the actual application capability of the proposed model.

In subsequent work, tool-wear state classification experiments under complex conditions will be conducted to further evaluate the performance of the HTT-Net model. The interpretability of the model will be improved, and the black-box limitations of deep learning will be reduced to promote the application of the model in industry. Furthermore, the performance of the HTT-Net model will be upgraded by integrating deep learning methods, such as data distillation.

CRediT authorship contribution statement

Zhongling Xue: Conceptualization, Methodology, Software, Validation, Data curation, Writing – original draft, Writing – review & editing. **Ni Chen:** Conceptualization, Supervision, Funding acquisition. **Youling Wu:** Investigation, Validation. **Yinfei Yang:** Conceptualization, Supervision. **Liang Li:** Conceptualization, Methodology, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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